

NOTES FOR MATH 3150, WEEK 3

1. THE ABSOLUTE VALUE

Define

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Proposition 1. *We have the following properties:*

- (1) $|xy| = |x||y|$
- (2) $|x| \leq a$ if and only if $-a \leq x \leq a$.
- (3) $|x + y| \leq |x| + |y|$.

Proof. Properties (1) and (2) follow straight from definition; we focus on (3). By property (2) we have, as $|x| \leq a$ for $a = |x|$,

$$-|x| \leq x \leq |x|.$$

Similarly,

$$-|y| \leq y \leq |y|.$$

Therefore, we get

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

Again by property (2), we get that

$$|x + y| \leq |x| + |y|.$$

□

Remark. Property (3) above is called *the triangle inequality* and will be extensively used in this course.

Example: Solve the equation $|x - 3| + |x - 1| = 3$.

Proof. We have

$$|x - 1| = \begin{cases} x - 1 & \text{if } x \geq 1 \\ 1 - x & \text{if } x < 1 \end{cases}$$

and

$$|x - 3| = \begin{cases} x - 3 & \text{if } x \geq 3 \\ 3 - x & \text{if } x < 3 \end{cases}$$

We study several cases. First, assume that $x < 1$. In this case

$$|x - 1| = 1 - x \quad \text{and} \quad |x - 3| = 3 - x.$$

Then

$$\begin{aligned} 3 &= |x - 3| + |x - 1| \\ &= 3 - x + 1 - x \end{aligned}$$

which implies $x = \frac{1}{2}$. We have found one solution.

Now assume that $1 \leq x < 3$. In this case,

$$|x - 1| = x - 1 \quad \text{and} \quad |x - 3| = 3 - x.$$

Then

$$\begin{aligned} 3 &= |x - 3| + |x - 1| \\ &= 3 - x + x - 1 \\ &= 2 \end{aligned}$$

Hence, there are no solutions.

Finally, if $x \geq 3$, then

$$|x - 1| = x - 1 \quad \text{and} \quad |x - 3| = x - 3.$$

We get

$$\begin{aligned} 3 &= |x - 3| + |x - 1| \\ &= 2x - 4, \end{aligned}$$

which implies $x = \frac{7}{2}$. This is the second solution.

Summarizing, the solutions are $x = \frac{1}{2}$ and $x = \frac{7}{2}$. □

Sequences represent a fundamental tool in analysis. We define them below and introduce the concept of limit of a sequence.

2. SEQUENCES

A sequence is a function from $\mathbb{N}$ to $\mathbb{R}$, that is $x : \mathbb{N} \rightarrow \mathbb{R}$. Its values are denoted as

$$x_n = x(n) \in \mathbb{R}.$$

A sequence is denoted with $(x_n)_{n \geq 1}$, or simply $(x_n)_n$ when the initial value of n is not important.

Examples:

1. The sequence $(x_n)_{n \geq 1}$ defined by $x_n = \frac{1}{n}$.
2. The sequence $(y_n)_{n \geq 1}$ defined by $y_n = \frac{(-1)^n}{2n}$.

In calculus we learned an intuitive, informal definition of limit. Now we give a formal, rigorous one. **This is a key concept in this subject.**

Definition 2. A sequence $(x_n)_{n \geq 1}$ is said to converge to a limit $x \in \mathbb{R}$ if for any $\varepsilon > 0$ there exists $N > 0$ so that

$$|x_n - x| < \varepsilon$$

for all $n \geq N$. In this case, we denote

$$x = \lim_{n \rightarrow \infty} x_n.$$

Note that N usually depends on ε , that is, the smaller ε is, the larger N needs to be.

Examples:

1. Prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0.$$

Proof. Fix any $\varepsilon > 0$. We want to find N so that

$$\frac{1}{n^2} < \varepsilon$$

for all $n > N$.

Solving this inequality we get $n > \frac{1}{\sqrt{\varepsilon}}$.

So we may just take $N = \frac{1}{\sqrt{\varepsilon}}$, which guarantees that $\frac{1}{n^2} < \varepsilon$ for all $n \geq N$. □

2. Prove that

$$\lim_{n \rightarrow \infty} \frac{2n-1}{n+1} = 2.$$

Proof. For any fixed $\varepsilon > 0$, we want that

$$\left| \frac{2n-1}{n+1} - 2 \right| < \varepsilon$$

for all $n > N$ large enough.

Calculating directly we get

$$\left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1}.$$

But $\frac{3}{n+1} < \varepsilon$ if and only if $n > \frac{3}{\varepsilon} - 1$.

So we may just take $N = \frac{3}{\varepsilon}$, then

$$\left| \frac{2n-1}{n+1} - 2 \right| < \varepsilon$$

for all $n > N$.

This proves that

$$\lim_{n \rightarrow \infty} \frac{2n-1}{n+1} = 2.$$

□

3. Prove that

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n^2 + n}}{n} = 1.$$

Proof. Fix any $\varepsilon > 0$. We want to prove that

$$\left| \frac{\sqrt{n^2 + n}}{n} - 1 \right| < \varepsilon$$

for all $n > N$ large enough.

We have

$$\left| \frac{\sqrt{n^2 + n}}{n} - 1 \right| = \left| \frac{\sqrt{n^2 + n} - n}{n} \right|.$$

When n is large, $\sqrt{n^2 + n}$ is also large, so we need a way to estimate the size of $\sqrt{n^2 + n} - n$. We introduce a very useful trick, that of multiplying with the conjugate.

That is, use that

$$\sqrt{a} - \sqrt{b} = \frac{a - b}{\sqrt{a} + \sqrt{b}}.$$

Apply this to $a = n^2 + n$ and $b = n^2$ we get

$$\begin{aligned} \sqrt{n^2 + n} - n &= \frac{n^2 + n - n^2}{\sqrt{n^2 + n} + n} \\ &= \frac{n}{\sqrt{n^2 + n} + n}. \end{aligned}$$

Therefore,

$$\left| \frac{\sqrt{n^2 + n} - n}{n} \right| = \frac{1}{\sqrt{n^2 + n} + n}.$$

We want to find a range of n that ensures that the right side is smaller than ε . We could solve it as an inequality, but there is a simpler way. We can further estimate

$$\frac{1}{\sqrt{n^2 + n} + n} < \frac{1}{n},$$

which implies that

$$\left| \frac{\sqrt{n^2 + n} - n}{n} \right| < \frac{1}{n}.$$

Let $N = \frac{1}{\varepsilon}$. It follows from above that

$$\left| \frac{\sqrt{n^2 + n} - n}{n} \right| < \frac{1}{n} < \varepsilon$$

for all $n > N$. This concludes the proof. $\square$

Not every sequence has a limit! For example, the sequence $(x_n)_n$ given by $x_n = (-1)^n$ does not have a limit.

Suppose it did, and let L be its limit. Then, for any $\varepsilon > 0$ there exists N so that

$$|(-1)^n - L| < \varepsilon$$

for all $n > N$.

Taking $n > N$ even implies $|L - 1| < \varepsilon$.

Taking $n > N$ odd implies $|L + 1| < \varepsilon$.

For $\varepsilon = \frac{1}{2}$, this is a contradiction.

3. PROPERTIES OF LIMITS

1. Limits are unique. If $x_n \rightarrow x$ and $x_n \rightarrow \bar{x}$, then $x = \bar{x}$.

Proof. By the limit definition, we have

$$|x_n - x| < \varepsilon$$

for all $n > N_1$, and

$$|x_n - \bar{x}| < \varepsilon,$$

for all $n > N_2$.

Adding them up, and using the triangle inequality, we have

$$\begin{aligned} |x - \bar{x}| &= |(x - x_n) + (x_n - \bar{x})| \\ &\leq |x - x_n| + |x_n - \bar{x}| \\ &\leq 2\varepsilon, \end{aligned}$$

for all $n > \max\{N_1, N_2\}$. Since ε is arbitrary, this proves $x = \bar{x}$. $\square$

2. Convergent sequences are bounded.

Proof. Assume that $(x_n)_n$ converges to x i.e., for any $\varepsilon > 0$ there exists $N > 0$ so that

$$|x_n - x| < \varepsilon$$

for all $n > N$. In particular, for $\varepsilon = 1$, it follows that $|x_n - x| < 1$ for all $n > N_1$. We rewrite this as

$$x - 1 < x_n < x + 1$$

for all $N > N_1$. Since N_1 is fixed number, this shows that $(x_n)_n$ is bounded. $\square$

3. Sum of limits. *If $x_n \rightarrow x$ and $y_n \rightarrow y$, then*

$$ax_n + by_n \rightarrow ax + by$$

for any $a, b \in \mathbb{R}$.

Proof. Fix any $\varepsilon > 0$. There exists $N_1, N_2 > 0$ so that

$$|x_n - x| < \varepsilon$$

for all $n > N_1$ and

$$|y_n - y| < \varepsilon$$

for all $n > N_2$.

It follows that for any $n > N = \max\{N_1, N_2\}$,

$$\begin{aligned} |(ax_n + by_n) - (ax + by)| &= |a(x_n - x) + b(y_n - y)| \\ &\leq |a||x_n - x| + |b||y_n - y| \\ &< (|a| + |b|)\varepsilon, \end{aligned}$$

where in the second line we used the triangle inequality, and in the third line we used that $|x_n - x| < \varepsilon$ and similarly for y_n .

Now we can finish the proof. For arbitrary $\epsilon > 0$ let $\varepsilon = \frac{\epsilon}{|a|+|b|+1}$. Accordingly, we have that

$$|(ax_n + by_n) - (ax + by)| < (|a| + |b| + 1)\varepsilon = \epsilon$$

for all $n > N$. This proves that $(ax_n + by_n)_n$ converges to $ax + by$. $\square$

4. Multiplication of limits. *If $x_n \rightarrow x$ and $y_n \rightarrow y$, then $x_n y_n \rightarrow xy$.*

Proof. The proof is similar to above and uses that

$$\begin{aligned} |x_n y_n - xy| &= |(x_n - x)y_n + x(y_n - y)| \\ &\leq |x_n - x||y_n| + |x||y_n - y|. \end{aligned}$$

We try to repeat the previous argument. An important difference is that $|y_n|$ is not a fixed number anymore. We cannot imitate the previous proof and define $\varepsilon = \frac{\epsilon}{|y_n|+|x|+1}$, because the right side changes with n .

The solution is to use that converging sequences are bounded. Therefore, there exists $M > 0$ such that $|y_n| < M$ for all n . The above inequality now says that

$$|x_n y_n - xy| \leq |x_n - x||y_n| + |x||y_n - y| \leq M|x_n - x| + |x||y_n - y|.$$

For any $\varepsilon > 0$ there exists $N_1 > 0$ such that

$$|x_n - x| < \varepsilon$$

for all $n > N_1$ and there exists $N_2 > 0$ such that

$$|y_n - y| < \varepsilon$$

for all $n > N_2$.

Therefore, for any $n > N = \max\{N_1, N_2\}$, we have

$$|x_n y_n - xy| < (M + |x|) \varepsilon.$$

We now finish as in the previous proof. For arbitrary $\epsilon > 0$ let $\varepsilon = \frac{\epsilon}{M+|x|+1}$. We conclude that

$$|x_n y_n - xy| < \epsilon.$$

for all $n > N$, which proves that $(x_n y_n)_n$ converges to xy . □

5. Quotient of limits. If $x_n \rightarrow x$ and $y_n \rightarrow y$ with $y \neq 0$, then $\frac{x_n}{y_n} \rightarrow \frac{x}{y}$.

The proof is very similar to the previous one, so we omit it.

6. The Squeeze Theorem. If $x_n \rightarrow L$, $y_n \rightarrow L$ and

$$x_n \leq z_n \leq y_n$$

for all $n \geq N$, then $z_n \rightarrow L$.

Proof. Fix any $\varepsilon > 0$. Since $x_n \rightarrow L$ we have

$$|x_n - L| < \varepsilon$$

for all $n > N_1$. Since $y_n \rightarrow L$, we have

$$|y_n - L| < \varepsilon$$

for all $n > N_2$.

In particular,

$$x_n > L - \varepsilon \text{ and } y_n < L + \varepsilon$$

for all $n > \max\{N_1, N_2\}$.

We conclude that

$$z_n \leq y_n < L + \varepsilon$$

and

$$z_n \geq x_n > L - \varepsilon$$

for all $n > \max\{N_1, N_2\}$.

This means that

$$|z_n - L| < \varepsilon$$

for all $n > \max\{N_1, N_2\}$. This proves the result. □